



Master 2 Internship report

Dynamical properties of the Hopfield model through bitwise arithmetic

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Abstract

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1 Introduction

1.1 Presentation of the host laboratories

This internship was conducted as part of my Master 2 at the Magistère of Fundamental Physics of Paris-Saclay Université, which I completed at Aix-Marseille Université in the “Physique Fondamentale et Applications” track. It was carried out at the “Physique des Cellules et Cancer” (PCC, UMR 168) laboratory of the Institut Curie as well as at the Gulliver laboratory (UMR 7083) of the “École Supérieure de Physique et de Chimie Industrielles” (ESPCI) in Paris, and was supervised by Michele Castellana (Institut Curie) and David Lacoste (ESPCI). Although the PCC unit is a leading biological physics laboratory, part of the Institut Curie – a pioneer institution in cancer research and treatment – it hosts a renowned theory team (of which Michele Castellana is a member) that is sometimes interested in physics-driven topics such as the one investigated in this report. The Gulliver laboratory hosts both experimentalists and theoreticians (among whom David Lacoste), working at the frontier between physical, chemical and biological topics.

1.2 Subject

In October 2024, John J. Hopfield was awarded the Nobel Prize in Physics alongside Geoffrey Hinton for their pioneering work on artificial neural networks [1]. Their work laid the foundations for further development and understanding of neural networks, whether they are organic or artificial, and eventually gave rise to Artificial Intelligence (AI).

Hopfield worked on an artificial neural network, originally invented by Shun’ichi Amari in 1972 [2]. He greatly contributed to the analysis, understanding and popularization of the network in his landmark paper *Neural networks and physical systems with emergent collective computational abilities* published in 1982 [3], and the network was eventually named after him. The model, described in more details in subsection ??, is similar to the Ising and the Spin Glass networks: it is composed of a set of spins – called neurons in the case of the Hopfield network – $-S_i \in \{-1, 1\}$ interacting with one another. In the same way as the two preceding models, the Hopfield model is energy-based, that is it will tend to minimize its energy over time. What makes this model different from the other two is the way the interactions between neurons are defined: as for the Ising and Spin Glass models they remain static over time but depend on the set of patterns to be stored in the network, making the latter attractors – minima of energy – of the model.

The Hopfield model provided the first simple yet powerful framework for understanding networks of interconnected neurons. It has become a reference in statistical physics on networks and its properties at equilibrium have been extensively studied and are now well known, whether it is its statistical mechanics [4, 5, 6] or its phase diagram and transitions [7]. However, much less is known about the dynamics that lead to those well-known equilibrium properties.

1.3 Objectives

The main objective of this internship is to efficiently simulate the Metropolis dynamics on the Hopfield network. To this end, we use a bitwise formalism that allows us to simulate several replicas of the system in parallel, increasing the efficiency of the exploration of the model’s energy landscape. Our objectives in this internship are thus twofold: first, to validate the bitwise implementation and quantify its computational gain in yield over the standard Metropolis algorithm; second, to leverage this implementation to study the dynamics of the Hopfield network. To validate the implementation, we begin with the Ising model, which serves as a natural benchmark due to its simplicity and well-characterized phase transition. Once this implementation has been proven sane and efficient on the Ising network, we increase the complexity by implementing it on the Spin Glass network, testing the gain in yield again.

Once the bitwise arithmetic has been validated on both models, we implement it on the Hopfield network in order to analyze its dynamics.

2 Implementation of the algorithm

We start by presenting the broad implementation of the Metropolis algorithm, before showing how we adapt it to each specific model.

2.1 The Metropolis algorithm

In order to make the models evolve in time, which is necessary to observe their dynamics, we use the well-known metropolis algorithm [8]. For models with a defined Hamiltonian, the Metropolis algorithm proceeds as follows:

1. Start from an initial configuration $\{S_i\}$.
2. Repeat N times:
 - a. Pick a random spin S_k and compute the energy change ΔE_k upon flipping it.
 - b. If $\Delta E_k \leq 0$, accept the flip.
 - c. Else, draw a random number $r \sim \mathcal{U}([0, 1])$ and accept the flip if $r \leq e^{-\beta \Delta E_k}$.
3. Repeat Step 2 for N_{sweeps} sweeps.

Where N is the number of spins, β is the inverse temperature, and N_{sweeps} is the number of sweeps, where one sweep consists of N successive single-spin flip attempts.

This algorithm however becomes computationally demanding when one attempts to sample a large number of configurations. This situation occurs, for example, when looking for rare samples in the dynamics, or sampling many different initial conditions – both of which we will try to explore during this internship. We thus need to find way of optimizing the implementation of the algorithm.

2.2 First optimization: the bitwise arithmetic

The first optimization in the implementation of the Metropolis algorithm is the use of bitwise arithmetic. The main idea is to leverage the boolean representation as stored in our computers. The typical object that we manipulate is called **Bits**: it is a collection of $n_{\text{bits}} = 64$ boolean variables. In that way, we have the possibility to simulate 64 evolutions of the same variable in parallel.

$$W = \underbrace{\boxed{b^0} \boxed{b^1} \boxed{b^2} \cdots \boxed{b^{62}} \boxed{b^{63}}}_{64 \text{ bits} \leftrightarrow 64 \text{ replicas}}$$

Figure 1: A **Bits** W encoding 64 parallel replicas. Each bit b_i carries the state of replica i .

This structure is very convenient to represent binary variable such as the spins of the system. The boolean representation $b_i \in \{0, 1\}$ of a spin $s_i \in \{-1, +1\}$ is given by $b_i = (s_i + 1)/2$. In that way, we are able to implement several replicas of the full spin system in a set of **Bits** $\{S_i\}$, each encoding the replicas of the associated spin, as shown in Figure 2.

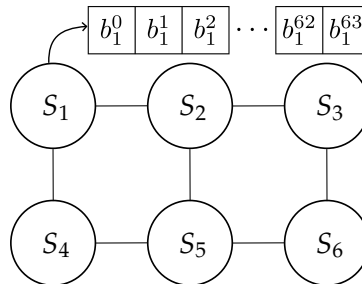


Figure 2: Bitwise implementation of a spin system. Each replica of the system is obtained by selecting the corresponding replica in each spin.

Storing all replicas in a single `Bits` object allows us to manipulate and update them simultaneously using a single bitwise operator. This is an instance of SWAR (SIMD Within A Register) parallelism, where SIMD stands for Single Instruction, Multiple Data: all 64 replicas are updated by a single bitwise instruction. The restriction to $n_{\text{bits}} = 64$ reflects the native word size of modern 64-bit CPUs [9]. This could in principle be extended to 512 bits using AVX-512 SIMD instructions, at the cost of introducing a new register type in the implementation. This will be left for further discussion. The convention used for what follows is presented in the Table 1:

Object	Notation
Scalar spin	$s_i \in \{-1, +1\}$
Scalar bit	$b_i \in \{0, 1\}$
64 replicas of spin i	$S_i \in \{-1, 1\}^{64}$
64-bit word (<code>Bits</code>)	$B_i \in \{0, 1\}^{64}$
Replica k of spin i	$s_i^{(k)} \in \{-1, +1\}$
Replica k of bit i	$b_i^{(k)} \in \{0, 1\}$

Table 1: Notation for the bitwise representation of spins.

Now that we know how to conveniently handle binary variables such as spins, we need to employ the same formalism to represent integer, such as the sum ΔE_k . This is done using a `BitSet`, which is a collection of `Bits`. When each `Bits` is associated with a power of 2, the `BitSet` represent a collection of n_{bits} unsigned integers and the `BitSet` is called `UnsignedInt`. We show an example of `UnsignedInt` in the Figure 3.

$$\begin{array}{c}
 \begin{array}{|c|c|c|} \hline b_0^0 & b_0^1 & b_0^2 \\ \hline \end{array} \cdots \begin{array}{|c|c|} \hline b_0^{62} & b_0^{63} \\ \hline \end{array} 2^0 \\
 \begin{array}{|c|c|c|} \hline b_1^0 & b_1^1 & b_1^2 \\ \hline \end{array} \cdots \begin{array}{|c|c|} \hline b_1^{62} & b_1^{63} \\ \hline \end{array} 2^1 \\
 \vdots \qquad \qquad \qquad \vdots \\
 \begin{array}{|c|c|c|} \hline b_l^0 & b_l^1 & b_l^2 \\ \hline \end{array} \cdots \begin{array}{|c|c|} \hline b_l^{62} & b_l^{63} \\ \hline \end{array} 2^l \\
 \underbrace{\hspace{10em}}_{64 \text{ replicas}}
 \end{array}
 \quad Z =$$

Figure 3: An `UnsignedInt` encoding 64 parallel unsigned integers. Each row is a `Bits` associated with a power of 2. The value of each integer is obtained by reading the `UnsignedInt` along the corresponding column. For instance, the fifth integer stored in Z is $Z_4 = \sum_{i=0}^l b_i^4 2^i$.

In the same way as for `Bits`, the `UnsignedInt` data structure enables us to add, subtract or even multiply two sets of 64 integers with just the cost of one operation. This implementation method has proven efficient when dealing with discrete values models such as the Ising [10] or the Spin Glass [11] models, but it has not yet been employed to study the Hopfield network.

2.3 Second Optimization: Shared random numbers

The bitwise implementation is a structural optimization of the data representation. On top of this structural optimization can be added a physical one which consists in having all replicas share the same sequence of random numbers, fully parallelizing their evolution. This turns out to be central: random number generation is often the simulation bottleneck, as it involves costly operations such as multiplications and is called at every step of the hot loop. A schematic view of the shared random number is shown in Figure 4. Using the same random number among all evolutions raises the questions of the independence of the realizations, this will be the subject of the next subsection.

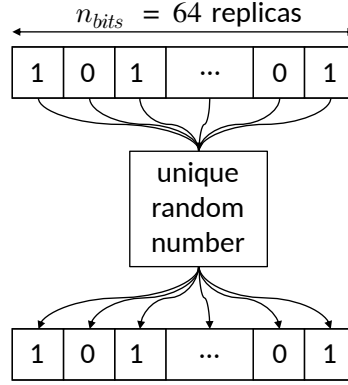


Figure 4: All 64 replicas of the system share the same random number at each step. **TO BE CHANGED (BOTTOW ROW)**

2.4 Note on the independence of the realizations

2.5 Structure of the code

The simulation of the dynamics has been entirely coded in C++, using an object-oriented approach to the problem.

$$S_1 = \begin{bmatrix} b_1^0 & b_1^1 & b_1^2 & \dots & b_1^{62} & b_1^{63} \end{bmatrix}$$

2.6 Standard implementation of the Metropolis algorithm

A first naive implementation of the classical algorithm is given in the Algorithm 3 in the appendix. A more efficient implementation would be to share the drawn random threshold across the replicas, this implementation is shown in the Algorithm 1:

Algorithm 1: Ising Metropolis sweep (classical)

Input: Lattice spins $\{S_i\}$, coupling J , inverse temperature β

Output: Updated lattice spins $\{S_i\}$

```

for sweep = 1 to  $N_{\text{sweeps}}$  do
  for  $k = 1$  to  $N$  do
    Draw  $\lfloor \rho \rfloor$  with  $\rho \sim \mathcal{E}(2\beta J)$ 
    for  $r = 1$  to  $n_{\text{bits}}$  do
      Compute  $\Lambda_k = s_k^{(r)} \sum_{i \in \partial k} s_i^{(r)}$ 
      // Unconditional flip: energy is lowered
      if  $\Lambda_k^{(r)} \leq 0$  then
         $s_k^{(r)} \leftarrow -s_k^{(r)}$ 
      else
        // Conditional flip
        if  $\lfloor \rho \rfloor \geq \Lambda_k^{(r)}$  then
           $s_k^{(r)} \leftarrow -s_k^{(r)}$ 

```

This reduces the number of random draws per n_{bits} cycle update to just one, shared amongst all replicas. This optimization is possible even though the structure is not fully parallelized across replicas. The only case for which this parallelized random number optimization might not be as efficient as expected is when $\Lambda_k^{(r)} \leq 0$ for all replicas simultaneously, in which case a random threshold is drawn

but never used. This standard or reference implementation will be broadly shared across all three models studied in this report.

3 The Ising model

We first implement the method on the Ising model, which is the simplest one.

3.1 The model

The Hamiltonian of the Ising network is given by

$$H_{\text{Ising}} = -J \sum_{\langle i,j \rangle} S_i S_j - h \sum_i S_i,$$

which, in absence of external field h becomes,

$$H_{\text{Ising}, h=0} = -J \sum_{\langle i,j \rangle} S_i S_j. \quad (1)$$

When flipping the spin S_k ($S_k \rightarrow -S_k$), the associated energy shift is

$$\begin{aligned} \Delta E_k &= H_{\text{Ising}}[-S_k] - H_{\text{Ising}}[S_k] \\ &= 2JS_k \sum_{i \in \partial k} S_i. \end{aligned} \quad (2)$$

where $i \in \partial k$ denotes the summation over the neighbors of k . If $\Delta E_k \leq 0$ the flip is accepted. Else, it is accepted if

$$\begin{aligned} r &\leq e^{-\beta \Delta E_k} \\ \Rightarrow \frac{-1}{\beta} \log r &\geq \Delta E_k \\ \Rightarrow \frac{-1}{2\beta J} \log r &\geq S_k \sum_{i \in \partial k} S_i. \end{aligned} \quad (3)$$

With $r \sim \mathcal{U}([0, 1])$, we know that $-\log r \sim \mathcal{E}(1)$ [[source?](#)], thus $-(2\beta J)^{-1} \log r = \rho \sim \mathcal{E}(2\beta J)$. The left-hand side is a random variable of probability density $p(\rho) = 2\beta J e^{-2\beta J \rho}$ and mean $(2\beta J)^{-1}$. The right-hand side of the equation is an integer which can be positive or negative. (However, due to the first escape condition of the algorithm, we know that it is strictly positive.) The left-hand side is thus compared with an (positive) integer. We can thus only consider the whole part of it without changing the inequality in order to obtain an integer-only flip condition:

$$\lfloor \rho \rfloor \geq S_k \sum_{i \in \partial k} S_i. \quad (4)$$

We now need to write the sum on spins in a Bits formalism in order to properly parallelize the evolutions.

3.2 Bitwise Implementation

Remembering that S_i is actually an array containing 64 replicas of the same spin, we write $S_i = -1 + 2B_i$, with B_i a Bits. The sum hence becomes:

$$\begin{aligned} \sum_{i \in \partial k} S_k S_i &= \sum_{i \in \partial k} (-1 + 2B_k)(-1 + 2B_i) \\ &= \sum_{i \in \partial k} 1 - 2(B_k + B_i) + 4B_k \cdot B_i, \end{aligned} \quad (5)$$

We show in Table 2 that the term in the sum can be re-written using the XNOR ($\odot$) logical operator. Thus, the term in the sum is simply equal to $2B_k \odot B_i - 1$, where $\odot$ denotes the bitwise XNOR operation

b_k	b_i	$1 - 2(b_k + b_i) + 4b_k \cdot b_i$	$b_k \odot b_i$	$2b_k \odot b_i - 1$
0	0	1	1	1
0	1	-1	0	-1
1	0	-1	0	-1
1	1	1	1	1

Table 2: The summed term can be simplified using a XNOR operator.

applied to the corresponding bits of each `Bits`. The sum thus becomes:

$$\sum_{i \in \partial k} S_k S_i = 2 \sum_{i \in \partial k} C_{ki} - n_k, \quad (6)$$

with $C_{ki} = 2B_k \odot B_i$ and n_k the number of neighbor of the spin k . The bitwise flip condition is then:

$$\lfloor \rho \rfloor + n_k \geq 2 \sum_{i \in \partial k} C_{ki}. \quad (7)$$

In addition, according to the definition of ΔE_k (2), we have $-2Jn_k \leq \Delta E_k \leq 2Jn_k$. Plugging this into Eq (3) yields the flip escape condition:

$$\lfloor \rho \rfloor \geq n_k. \quad (8)$$

When satisfied, this guarantees that Eq. (4) holds for every replica regardless of its local configuration, so that all replicas are flipped simultaneously without computing the per-replica bitwise test. This first escape is useful, as it avoids the bitwise flipping test calculations for each replica, even when done in parallel. Instead, a simple integer comparison suffices. This condition is typically met in the high-temperature regime. Moreover, the integer flip condition (8) constrains the range of values taken by $\lfloor \rho \rfloor$ and $2 \sum_{i \in \partial k} C_{ki}$ to n_k and $2n_k$, respectively. This is particularly convenient, as it allows the `UnsignedInt` objects to be allocated with static sizes at initialization, removing the need for dynamic resizing during the simulation.

The pseudocode for the bitwise implementation of the Metropolis algorithm on this Ising network is presented in the algorithm 2.

Algorithm 2: Ising Metropolis sweep (bitwise)

Input: Lattice spins $\{S_i\}$, coupling J , inverse temperature β

Output: Updated lattice spins $\{S_i\}$

Variables: Bits `xnor`, `mask`

UnsignedInt T_k , threshold**for** $sweep = 1$ **to** N_{sweeps} **do****for** $step = 1$ **to** N **do**

Draw $\lfloor \rho \rfloor$ with $\rho \sim \mathcal{E}(2\beta J)$

if $\lfloor \rho \rfloor \geq n_k$ then

```

 $S_k \leftarrow -S_k$  // escape condition: unconditional flip across all replicas

```

else

$$T_k \leftarrow \mathbf{0}$$
$$\text{threshold} \leftarrow 0$$
for $i \in \partial k$ **do**
$$\text{xnor} \leftarrow B_k \odot B_i$$
$$T_k \mathrel{+}= \text{xnor}$$

```
// bitwise XNOR across replicas
```

```
// increment agreement count per replica
```

$$T_k \leftarrow 2T_k$$
$$\text{threshold} \leftarrow \lfloor \rho \rfloor + n_k$$
$$\text{mask} \leftarrow (T_k \leq \text{threshold})$$

```
// per-replica flip condition
```

$$B_k \leftarrow B_k \oplus \text{mask}$$

```
// bitwise spin flip (XOR operation): only replicas
```

```
// for which  $\text{mask}^{(\ell)} = 1$  are flipped
```

3.3 Phase diagram

We now need to make sure the bitwise implementation is consistent. This is firstly done by checking that it yields exactly the same spin configurations as the standard implementation with shared random numbers when starting from the exact same initial condition and using the same generative seed. Secondly, we can plot the phase diagram of the model and verify that it is in agreement with the known analytical results. We show in Figure 5 the obtained phase diagram for several sizes of two-dimensional square lattices. This phase diagram is obtained by successive cooling the system: starting from the highest temperature, we progressively decrease it, using the final state of each step as the initial configuration for the next. This reduces the computational cost compared to reinitializing the network randomly at each temperature point, since the equilibrium state at each temperature lies close to that of the previous one, requiring fewer sweeps to reach thermalization.

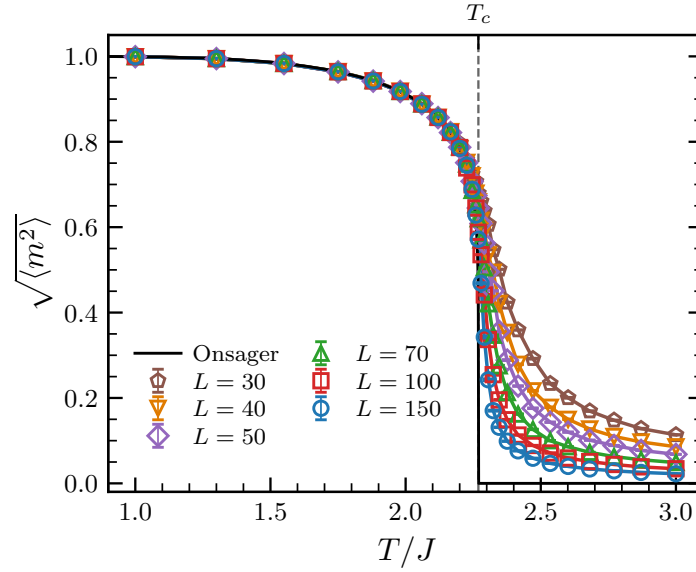


Figure 5: Phase diagram of the Ising model for several sizes of square lattices. The network is built with Periodic Boundary Conditions (PBC). The black line corresponds to the exact analytical solution found by Onsager in the case of an infinite square lattice [12]: $m(T) = 0$ for $T \geq T_c$ and $m(T)_{\pm} = \pm [1 - \sinh^{-4}(2K)]^{1/8}$ for $T \leq T_c$, with $K = J/T$ and $T_c = 2J/\log(1 + \sqrt{2}) \approx 2.269$.

Qualitatively, it appears that the magnetization curve gets closer to the exact solution when the lattice size is increased. This was to be expected as we gradually get closer to the infinite size case when increasing the lattice size. We also notice an excellent agreement between the exact analytical magnetization and the obtained ones in the ferromagnetic part of the phase diagram. Thus, the obtained phase diagram is in excellent agreement with the expected phase diagram, taking into account that we work on finite size lattices. In this plot, each point corresponds to the mean over the $n_{\text{bits}} = 64$ independent realizations. We show in Figure 6 the trajectories over time of the magnetizations for those 64 realizations.

In order to make sure that the phase transition of our network is well at $T_c = 2.269$ for all lattice sizes, we plot the Binder U_L , defined as:

$$U_L = 1 - \frac{\langle m^4 \rangle_L}{3\langle m^2 \rangle_L^2} \quad (9)$$

Its limit values are $U_L(0) = 2/3$ and $U_L(T \rightarrow \infty) = 0$. Most importantly, it has been shown that all Binder curves for different lattice sizes intersect at the same scale-invariant critical point (T_c, U^*) [13]. This is because at T_c , the system becomes scale invariant, making the Binder cumulant independent of the system size at this specific temperature. We show in Figure 7 the plot of the Binder cumulants for the different lattice sizes.

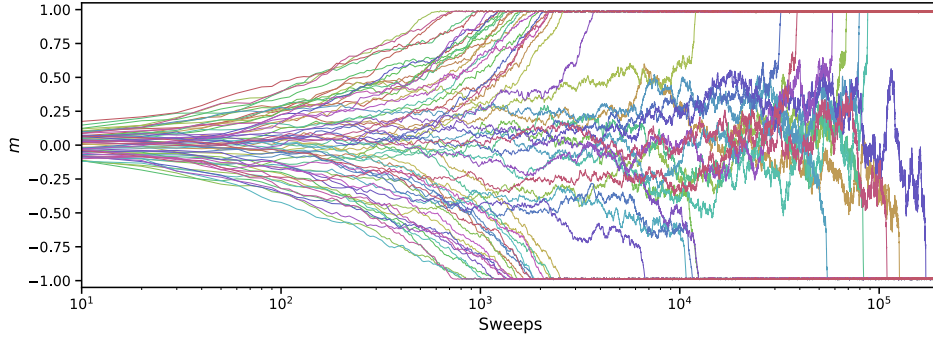


Figure 6: Trajectories of the 64 magnetizations over time for $L = 100$ and $T/J = 1.5$. All those trajectories have been obtained from the same simulation. While some realizations quickly collapse into the equilibrium attractors $m_{\pm} = \pm 0.953$, some require $\sim 10^2$ times more sweeps to reach the same equilibrium.

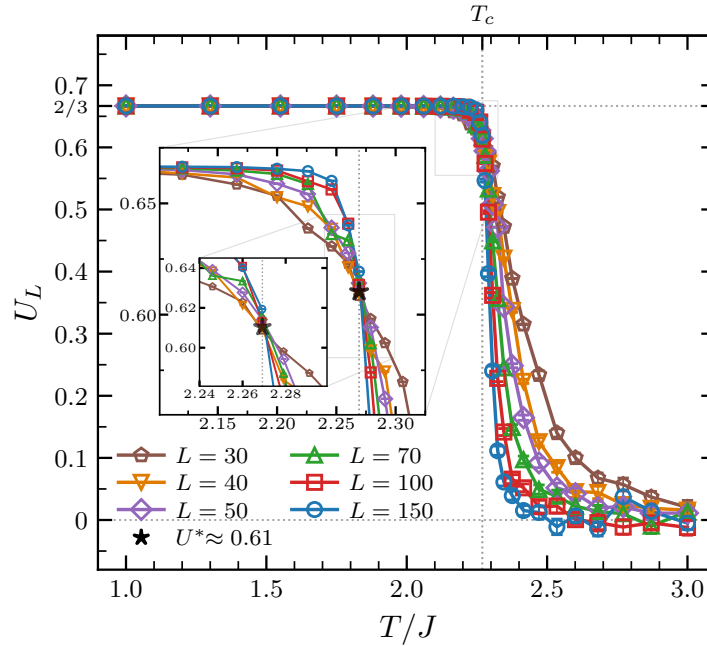
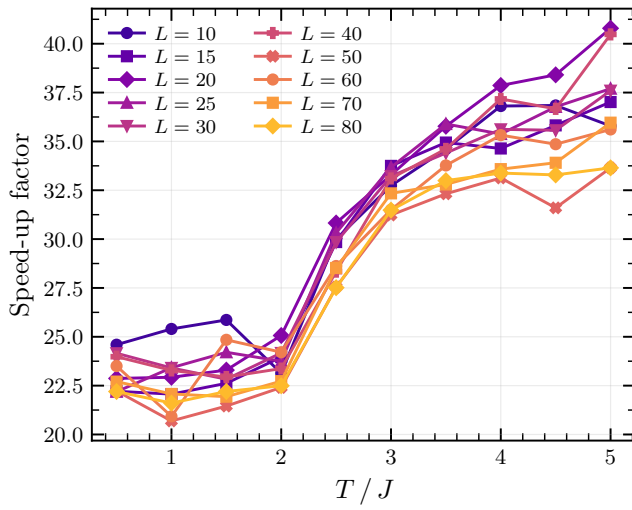


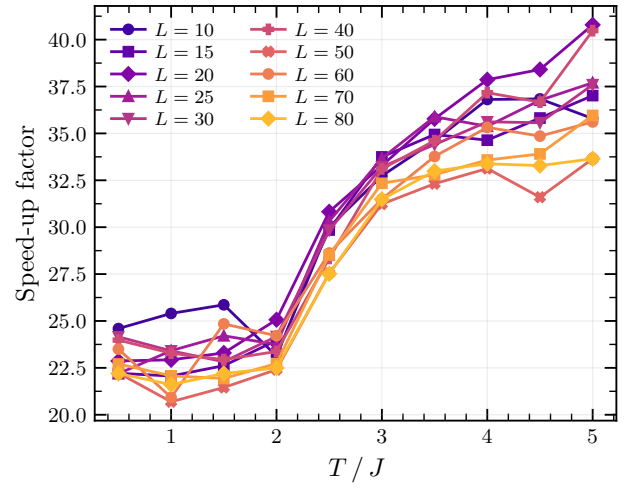
Figure 7: Binder cumulant for the different lattice sizes. The curves approach the two asymptotic limits and intersect at (T_c, U^*) , as expected. Insets show the zoomed region around the critical point.

3.4 Performance test

Now that our algorithm has been shown to produce correct results, we wish to estimate its performance compared to the usual naive implementation. To this end, we measure the ratio of the computational time required to evolve the system over $N_{\text{sweeps}} = 2^{16}$ sweeps using both the classical naive implementation and the bitwise implementation, for various square lattice sizes and temperatures. The result of this performance test is shown in Figure 8. The speedup factor lies between 20 and 40, which is a substantial increase. We notice that this speedup factor consistently increases for the highest temperatures. Surprisingly, that is not so much because of the escape condition in the bitwise condition, which is more likely at highest temperatures. It is mainly because at low temperatures, the system quickly thermalizes to an ordered configuration where spins are aligned with their neighbors, making the unconditional flip condition $\Delta E_k \leq 0$ more frequent in the classical implementation, which is computationally cheaper. The increases however seems to be uncorrelated with the size of the system. In theory, this factor can even be larger than 64, as the structural optimization alone can potentially yield a speedup of 64 by itself, as it enables us to make 64 realizations evolve in parallel. However, this structural optimization comes with the cost of creating



(a) Speedup factor heatmap.



(b) Time per system size and number of sweeps, averaged over all system sizes, for different temperatures.

Figure 8: Speedup factor between classical and bitwise implementations of the Ising Metropolis algorithm for several lattice sizes and temperatures.

and storing the bitwise objects used to parallelize the evolution, thus reducing the net performance improvement.

It must be noted that, while we here measure an increase in the speed of the code for a fixed number of outputs, the true advantage of the bitwise implementation is to produce more independent results within the same computational time.

4 The Spinglass model

We now move on a more complex model, in order to approach the structure of the Hopfield network.

4.1 The Edwards–Anderson model

The Edwards–Anderson (EA) model – first introduced by Edwards and Anderson in 1975 [14] – is a finite dimensional spin glass model with short range interaction. It is not a mean field spin glass model with infinite range interactions such as the Sherrington–Kirkpatrick model [15]. The model is formally described as follows:

$$H_{EA} = \sum_{\langle i,j \rangle} J_{ij} S_i S_j. \quad (10)$$

Where J_{ij} is a random variable taking its values in $\{-1, 1\}$ with equal probability. Thus, it can easily be represented by a Bits. Under a spin flip ($S_k \rightarrow -S_k$), the energy shift is:

$$\Delta E_k = 2S_k \sum_{i \in \partial k} J_{ki} S_i. \quad (11)$$

The structure being really similar to that of (2), we can perform the same derivation as for the Ising model and find the spin flip condition:

$$\lfloor \rho \rfloor \geq S_k \sum_{i \in \partial k} J_{ki} S_i, \quad (12)$$

with this time $\rho \sim \mathcal{E}(2\beta)$.

4.2 Bitwise Implementation

We now need to express the product $J_{ik} S_k S_i$ in a binary way. As shown in the previous section, the product $S_k S_i$ can be re-written $(-1 + 2B_k)(-1 + 2B_i) = -1 + 2C_{ki}$ with $C_{ki} = B_k \odot B_i$. Thus, by writing $J_{ki} = -1 + 2K_{ki}$ with K_{ki} a Bits, we have:

$$\begin{aligned} J_{ik} S_k S_i &= (-1 + 2K_{ki})(-1 + 2C_{ki}) \\ &= -1 + 2D_{ki}, \end{aligned} \quad (13)$$

with $D_{ki} = K_{ki} \odot C_{ki} = K_{ki} \odot B_k \odot B_i = K_{ki} \oplus B_k \oplus B_i$. As in the previous section, the flip condition is then:

$$\lfloor \rho \rfloor + n_k \geq 2 \sum_{i \in \partial k} D_{ki}. \quad (14)$$

Both the bitwise and the classic implementation are thus similar to this of the Ising model, and we will not present them further in this report. The bitwise algorithm for the spin glass model can be found in the appendix.

The bitwise implementation is not validated here, as the bitwise framework was already tested in the previous section, and the main interest of this model lies in bridging the complexity gap between the Ising model and the Hopfield network. We nonetheless verified that the standard and the bitwise implementations produce identical spin configurations when initialized from the same initial state and driven by the same sequence of random numbers.

4.3 Performance test

5 The Hopfield model

5.1 The model

$$H_{\text{Hopfield}} = - \sum_{i,j} w_{ij} S_i S_j \quad (15)$$

$$w_{ij} = \frac{1}{N} \sum_{\mu=1}^p \xi_i^\mu \xi_j^\mu \quad (16)$$

$$\begin{aligned} \Delta E_k &= 2 \sum_{i \neq k} w_{ki} S_k S_i \\ &= \frac{2}{N} \sum_{i \neq k} \sum_{\mu=1}^p \xi_k^\mu \xi_i^\mu S_k S_i \end{aligned} \quad (17)$$

5.2 Bitwise Implementation

We write $w_{ki} = \frac{1}{N} G_{ki}$ in order to deal with integer-only variables. Thus:

$$\frac{\Delta E_k N}{2} = \sum_{i \neq k} G_{ki} S_i S_k, \quad \text{with } G_{ki} \in [-p, p] \quad (18)$$

And the flip condition becomes:

$$\lfloor \rho \rfloor \geq \sum_{i \neq k} G_{ki} S_k S_i, \quad (19)$$

with $\rho = -\frac{N}{2p} \log r$.

5.2.1 Naive implementation

A first naive implementation would be to unfold G_{ki} into its pattern components in order to deal with Bits only objects. The flip condition would then be:

$$\lfloor \rho \rfloor \geq \sum_{i \neq k} \sum_{\mu=1}^p \xi_k^\mu \xi_i^\mu S_k S_i \quad (20)$$

As seen in the previous sections, the product $\xi_k^\mu \xi_i^\mu S_k S_i$ can simply be expressed from $X_k^\mu \odot X_i^\mu \odot B_k \odot B_i$, where X_k^μ is the boolean variable associated with ξ_k^μ . This method presents the advantage to not require too much memory allocation to store the couplings G_{ki} as they are not stored before the evolution but computed on the fly in the hot loop. However, this method requires summing over the patterns, thus introducing an additional complexity of $O(p) = O(\alpha N)$ per spin update, which is done $N \cdot N_{\text{sweeps}}$ times.

5.2.2 Optimized implementation

A more efficient way to go would be to stay with the integer factor G_{ki} , which is suitably stored into an `UnsignedInt` in the bitwise implementation. We operate the shift $\tilde{G}_{ki} = p + G_{ki} \in [0, 2p]$, in order to deal with positive only integers, as required by the `UnsignedInt` format. Then:

$$\sum_{i \neq k} G_{ki} S_i S_k = \sum_{i \neq k} (\tilde{G}_{ki} - p)(2C_{ik} - 1) = - \sum_{i \neq k} \tilde{G}_{ki} + 2 \sum_{i \neq k} C_{ik} (\tilde{G}_{ki} - p) + n_k p, \quad (21)$$

with $C_{ki} = B_k \odot B_i$.

$$\lfloor \rho \rfloor + \sum_{\langle k,i \rangle} \tilde{G}_{ki} + 2p \sum_{\langle k,i \rangle} C_{ki} \geq p n_k + 2 \sum_{\langle k,i \rangle} C_{ki} \oplus \tilde{G}_{ki} \quad (22)$$

Where $C_{ki} \oplus \tilde{G}_{ik}$ consists in the XOR operator between C_{ki} and every Bits composing the UnsignedInt $\tilde{G}_{ki}$:

$$(C_{ki} \oplus \tilde{G}_{ki})_l = C_{ki} \oplus (\tilde{G}_{ki})_l, \quad (23)$$

where l denotes the l -th Bits (associated with the power 2^l). This is strictly equivalent to setting the column of $\tilde{G}_{ik}$ to 0 where C_{ki} is zero and leave it untouched else. Maximum value of R : pN_k

Maximum value of $\sum_{\langle k,i \rangle} g_{ik}$: $2pN_k$

Maximum value of $2p \sum_{\langle k,i \rangle} c_{ik}$: $2pN_k$

Maximum value left-hand side: $5pN_k$

Maximum value right-hand side: $pN_k(1 + 2 \cdot 2) = 5pN_k$

The Metropolis condition $r \leq e^{-\beta \Delta E_k}$ is equivalent to:

$$R := \frac{-N}{2\beta} \log r \geq \frac{\Delta E_k N}{2} \quad (24)$$

at max, $\Delta E_k N / 2 = pN_k$. So if $R \geq pN_k$, unconditional flip.

Else, after substitution:

5.3 Capacity of the Hopfield network

5.4 Performance test

6 Conclusion

Appendices

A Naive implementation of the classic Metropolis algorithm

Algorithm 3: Ising Metropolis sweep (classical, naive)

Input: Lattice spins $\{S_i\}$, coupling J , inverse temperature β

Output: Updated lattice spins $\{S_i\}$

```

for sweep = 1 to  $N_{\text{sweeps}}$  do
  for  $k = 1$  to  $N$  do
    for  $r = 1$  to  $n_{\text{bits}}$  do
      Compute  $\Lambda_k = s_k^{(r)} \sum_{i \in \partial k} s_i^{(r)}$ 
      // Unconditional flip: energy is lowered
      if  $\Lambda_k^{(r)} \leq 0$  then
         $s_k^{(r)} \leftarrow -s_k^{(r)}$ 
      else
        // Conditional flip: draw Poisson threshold
        Draw  $\lfloor \rho \rfloor$  with  $\rho \sim \mathcal{E}(2\beta J)$ 
        if  $\lfloor \rho \rfloor \geq \Lambda_k^{(r)}$  then
           $s_k^{(r)} \leftarrow -s_k^{(r)}$ 

```

B Bitwise implementation of the Metropolis algorithm for the spin glass model

Algorithm 4: Ising Metropolis sweep (bitwise)

Input: Lattice spins $\{S_i\}$, couplings J_{ki} , inverse temperature β

Output: Updated lattice spins $\{S_i\}$

Variables: Bits xnor , mask

UnsignedInt T_k , threshold

```

for sweep = 1 to  $N_{\text{sweeps}}$  do
  for step = 1 to  $N$  do
    Draw  $\lfloor \rho \rfloor$  with  $\rho \sim \mathcal{E}(2\beta)$ 
    if  $\lfloor \rho \rfloor \geq n_k$  then
       $S_k \leftarrow -S_k$  // escape condition: unconditional flip across all replicas
    else
       $T_k \leftarrow 0$ 
      threshold  $\leftarrow 0$ 
      for  $i \in \partial k$  do
         $\text{xnor} \leftarrow K_{ki} \oplus B_k \oplus B_i$  // bitwise XNOR across replicas
         $T_k += \text{xnor}$  // increment agreement count per replica
       $T_k \leftarrow 2T_k$ 
      threshold  $\leftarrow \lfloor \rho \rfloor + n_k$ 
      mask  $\leftarrow (T_k \leq \text{threshold})$  // per-replica flip condition
       $B_k \leftarrow B_k \oplus \text{mask}$  // bitwise spin flip (XOR operation): only replicas
                                // for which  $\text{mask}^{(\ell)} = 1$  are flipped

```

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